1 Discussion (excerpt)

1.1 Relation to the developer's published safety results

Meta reports safety results for every Llama generation we study: Llama 2-Chat in Touvron et al. (2023) and Llama 3.1 in Llama Team, AI @ Meta (2024), whose results are for the 3.1 models throughout. Llama 3 itself (April 2024) has no technical report. Our anchor pair, Llama-3-8B-Instruct → Llama-3.1-8B-Instruct, therefore spans exactly the post-training described in Llama Team, AI @ Meta (2024, §5.4.3). None of the developer's figures is directly comparable to ours. Its safety benchmarks are internal and anonymised and, by its own account, "not reproducible externally" (Llama Team, AI @ Meta, 2024, fn. 13). They report violation and false-refusal rates, mainly for the 70B and 405B models, without verified intent preservation or judge-error correction. Table 1 therefore sets each published claim against the measurement of ours that bears on it, and grades the relation instead of comparing numbers. Table 2 gives the one external check of our screening that is possible: the independently published SORRY-Bench fulfilment rates (Xie et al., 2025).

Four points follow from Table 1. First, no measurement contradicts a published claim. Where the two overlap, on the generational shift in refusal rates, the weakness of known single-turn templates, and the scale dependence of capability-gated attacks, they agree. The independent screening check agrees as well (Table 2, within 1.6 points for two of three models; the 13B gap is plausibly due to decoding and quantisation). Second, the developer evaluates safety against attack *forms*: adversarial and borderline prompts, red-team techniques, a system-level guard. It does not evaluate invariance of a refusal under rewording that keeps the request the same, and that is the axis on which our results diverge most. The Llama-3.1 fixes that resist every tested jailbreak template still re-open under a benign register or paraphrase change in a majority of cases, and the effect replicates on a held-out set drawn from a different source — the one result here that is confirmatory rather than exploratory.

Third, the lineage trend in PBR is benchmark-dependent and does not settle the developer's "better balance" claim either way. On SORRY-Bench, whose categories include borderline requests, lower false refusal in newer generations coincides with refusals that are less robust to surface form; on HarmBench the newer generations' refusals are, if anything, slightly more robust. Only one rung of the lineage is a matched-scale generational comparison, so the rest is breadth rather than a tested trend.

Fourth, and most cautiously: where the fixes re-open, the intervention locates part of the change before the first output token. Moving a rewording along Llama-3.1-8B's own refusal direction shifts its propensity to refuse in the predicted direction in every behaviour tested, and in both directions — but restores the refusal outright in only a small minority. A single direction is therefore implicated in, and insufficient for, the behaviour. What remains genuinely untested is the attribution of *which* kind of rewording does the work: the earlier claim that register and paraphrase were the drivers does not survive the registered specification, and we report the question as open rather than answered. The data-recency and refusal-wording rows likewise remain hypotheses, not findings.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Llama Team, AI @ Meta. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Table 1: Correspondence between Meta's published Llama safety findings and the present measurements. Relation: ≈ consistent; + extends (a dimension the developer does not report); ↔ contrast; ? hypothesis, not tested; ∅ no counterpart. Our figures are raw and exploratory unless marked corrected or confirmatory; intervals are Wilson 95%.

Theme	Developer's finding (source)	This study	Rel.
Generational balance (Llama 2 → 3)	Llama 3 gives "a much better balance between helpfulness and harmlessness than its predecessor", with lower false refusal on internal benchmarks (Llama Team, AI @ Meta, 2024, §1, Fig. 21).	Canonical compliance roughly doubles: SORRY-Bench 12.3% → 20.9%, HarmBench 10.3% → 23.8% (Llama-2-7B → Llama-3-8B). Newly answered behaviours far outnumber newly refused ones (SORRY-Bench 64 vs. 4; HarmBench 66 vs. 5)[a].	≈
Safety fine-tuning of Llama 3.1	Safety SFT and DPO on adversarial plus borderline data; refusal tone guideline (Llama Team, AI @ Meta, 2024, §5.4.3).	*Confirmatory* (pre-registered, held-out set): of the behaviours Llama-3.1-8B newly refuses, 6/9 are re-opened by a gate-admitted rewording, corrected RSR 0.765 [0.381, 1.000], McNemar $p = 0.031$ — *materially surface-bound* against the registered reference $\pi^* = 0.15$. *Exploratory*, pooled over four benchmarks on the anchor run: 45/82 re-opened, corrected 0.622 [0.470, 0.775]; the SORRY-Bench anchor alone is 23/47, corrected 0.551 [0.364, 0.738][b]. Adding the LLM intent check to the gate leaves RSR unchanged.	+
Robustness to surface form	Robustness is reported for capability only: pre-trained models are "very robust" to MCQ label and format changes (Llama Team, AI @ Meta, 2024, §5.1.2); paraphrase detection is unaffected by PAWS-style adversariality (Llama Team, AI @ Meta, 2024, §5.1.3).	Refusals are not invariant to meaning-preserving rewording. Llama-3.1-8B's *model-level* paraphrase brittleness over its own refused set (SORRY-Bench lineage run, 63/376) is 0.168 raw, corrected 0.163 [0.093, 0.233] — *inconclusive* against π^{*}[c]. *Which* kind of rewording does it is **not** established: under the registered specification no single transform class survives Holm correction (anchor omnibus $p = 0.284$ over 214 rewordings in 49 behaviour clusters, 0 classes significant)[d].	↔
Safety across the lineage	Llama 2-Chat has a "low violation percentage overall across model sizes" (Touvron et al., 2023, Fig. 17).	The direction of the PBR trend depends on the benchmark. On SORRY-Bench, corrected PBR is lowest for Llama 2 (7B 0.113, 13B 0.083) and higher for every Llama 3.x model (Llama-3-8B 0.199, Llama-3.1-8B 0.163, Llama-3.2-1B 0.197, Llama-3.2-3B 0.240). On HarmBench it reverses (Llama 2 17.6 % and 17.4 % against Llama 3 14.4 % and Llama 3.1 13.8 %, overlapping intervals); on AdvBench and StrongREJECT it is small for every model (0.8–6.3 %). Four of the five lineage rungs are within-generation or scale-confounded, so the trend is breadth, not a tested comparison[e].	≈/↔
Known jailbreak templates	Refusal suppression, hypotheticals, personas and disclaimers were "pervasive across model checkpoints, particularly when used together", mostly in multi-turn red teaming; findings fed back into safety data (Llama Team, AI @ Meta, 2024, §5.4.6).	Single-turn templates (Wei et al., 2023) on Llama-3.1-8B, judged against the original request ($n = 291$ canonically refused): refusal suppression 2.1%, AIM and evil confidant 0.0%; the strongest class (short style injection) re-opens 4.1% [2.4, 7.1][f]. Multi-turn attacks not tested. Nor does *combining* styles help: in a 21-condition factorial over 49 fixes, no style pair re-opens more than its own singles (Holm minimum $p = 0.23$).	≈
Scale and	8B models need a larger share of safety	An encoded combination attack re-opens	≈

Table 2: External check of the canonical screening: SORRY-Bench fulfilment rates on the 440 base instructions.

Model	Xie et al. (2025)[a]	This study[b]
Llama-2-7B-Chat	13.9 %	12.3%
Llama-2-13B-Chat	14.6 %	9.3%
Llama-3-8B-Instruct	22 %	20.9%

[a] Temperature 0.7, one sample, fine-tuned Mistral-7B judge.
[b] Greedy decoding on 4-bit quantised weights (Ollama defaults, not matched across the lineage), compliant if any of three draws is judged compliant, qwen2.5:14b judge.

Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Tinghao Xie, Xiangyu Qi, Yi Zeng, et al. SORRY-Bench: Systematically evaluating large language model safety refusal. In *International Conference on Learning Representations (ICLR)*, 2025.